Basics of Financial Accounting

Asset Worth & Depreciation

An **enterprise** is a large organization or business that carries out commercial, industrial, or professional activities to produce goods or services and earn profit.

It can also refer more generally to any planned project or business venture involving effort, investment, and risk.

Why Engineers need to understand F-A/C

With the growth in importance of small technology-based enterprises, many engineers have taken on broad **managerial responsibilities** that include financial accounting. Financial accounting is concerned with **recording and organizing** the financial data of businesses. The data cover both ***flows over time*, like revenues and expenses, and *levels***, like an enterprise's resources and the claims on those resources at a given date. Even engineers who do not have broad managerial responsibilities need to know the elements of financial accounting to understand the enterprises with which they work.

Why Assets Depreciate?

Engineering projects often involve an investment in equipment, buildings, or other assets that are put to productive use. As time passes, these assets lose value, or *depreciate*.

Use-related physical loss: As something is used, parts wear out. An automobile engine has a limited life span because the metal parts within it wear out. This is one reason why a car diminishes in value over time. Often, use-related physical loss is measured with respect to *units of production*, such as units produced, hours operated, or miles driven.

Why Assets Depreciate?

Time-related physical loss: Even if something is **not used**, there can be a physical loss over time. This can be due to environmental factors affecting the asset or to endogenous physical factors. For example, an unused car can rust and thus lose value over time. Time related physical loss is expressed in units of time, such as a 10-year-old car or a 40-year-old sewage treatment plant. endogenous : internal cause within an asset

Functional loss: Losses can occur without any physical changes. For example, a car can lose value over time because styles change so that it is no longer fashionable. Other examples of causes of loss of value include legislative changes, such as for pollution control or safety devices, and technical changes. Functional loss is usually expressed

Value of an Asset

Models of depreciation can be used to estimate the loss in value of an asset over time, and also to determine the remaining value of the asset at any point in time. This remaining value has several names, depending on the circumstances.

Market value is usually taken as **the actual value an asset can be sold for in an open market**. Of course, the only way to determine the actual market value for an asset is to sell it. Consequently, the term *market value* usually means **an *estimate* of the market value.** One way to make such an estimation is by using a depreciation model that reasonably captures the true loss in value of an asset.

Book value is the depreciated value of an asset for accounting purposes, as calculated with a depreciation model. The book value may be more or less than market value. The depreciation model used to arrive at a book value might be controlled by regulation for some purposes, such as taxation, or simply by the desirability of an easy calculation scheme. **There might be several different book values for the same asset, depending on the purpose and depreciation model applied**.

Scrap value can be **either the actual value of an asset at the end of its physical life (when it is broken up for the material value of its parts)** or an estimate of the scrap value calculated using a depreciation model.

Salvage value can be either the **actual value of an asset at the end of its useful life (when it is sold**) or an estimate of the salvage value calculated using a depreciation model.

A company that sells a particular type of web-indexing software has had two larger firms approach it for a possible buyout. The current value of the company, based on recent financial statements, is $4.5 million. The two bids were for $4 million and $7 million, respectively. Both bids were bona fide, meaning they were real offers. What is the market value of the company? What is its book value? BV= $4.5 M, MV = $7 M. question 1 soln in notes

A new industrial sewing machine costs in the range of $5000 to $10 000. Technological change in sewing machines does not occur very quickly, nor is there much change in the functional requirements of a sewing machine. A machine can operate well for many years with proper care and maintenance. Discuss the different reasons for depreciation and which you think would be most appropriate for a sewing machine. Use related Physical Depreciation q2 soln in notes

Communications network switches are changing dramatically in price and functionality as changes in technology occur in the communications industry. Prices drop frequently as more functionality and capacity are achieved. A switch only six months old will have depreciated since it was installed. Discuss the different reasons for depreciation and which you think would be most appropriate for this switch. Depreciation on account of Functional Loss q3 and q4

Ryan owns a five-hectare plot of land in the countryside. He has been planning to build a cottage on the site for some time, but has not been able to afford it yet. However, five years ago, he dug a pond to collect rainwater as a water supply for the cottage. It has never been used and is beginning to fill in with plant life and garbage that has been dumped there. Ryan realizes that his investment in the pond has depreciated in value since he dug it. Discuss the different reasons for depreciation and which you think would be most appropriate for the pond.

Why to measure Value of Assets?

It is desirable to be able to construct a good model of depreciation in order **to state a book value of an asset** for a variety of reasons:

1. In order to make many managerial decisions, it is necessary **to know the value of owned assets. For example, money may be borrowed using the firm's assets as collateral**. In order to demonstrate to the lender that there is security for the loan, a **credible estimate of the assets' value** must be made.

2. One needs an estimate of the value of owned assets **for planning purposes.** In order to decide whether to keep an asset or replace it, you have to be able to judge how much it is worth. More than that, you have to be able **to assess how much it will be worth at some time in the future**.

3. **Government tax legislation requires that taxes be paid on company profits.** Because there can be many ways of calculating profits, strict rules are made concerning how to calculate income and expenses. **These rules include a particular scheme for determining depreciation expenses.**

Why to measure Value of Assets?

Depreciation – Tax Shield

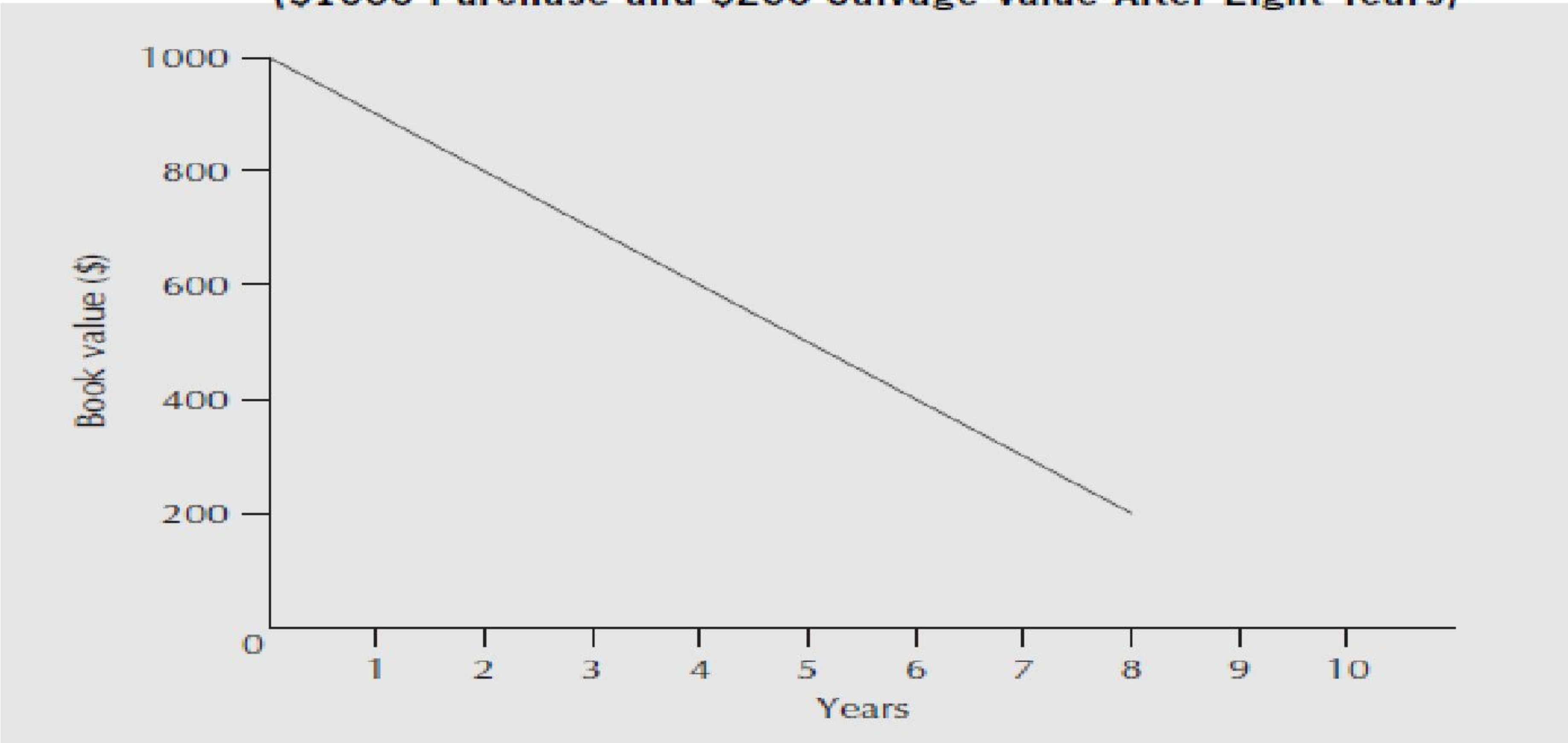
Depreciation acts as a **tax shield** because it is a non-cash expense that **reduces a company's taxable income**, allowing it to pay less in taxes. Because no cash actually leaves the company when depreciation is recorded, this reduction in tax creates a direct increase in cash flow.

Depreciation is called a tax shield because it reduces the amount of tax a company has to pay.

Straight-line method of depreciation

The **straight-line method of depreciation** assumes that the rate of loss in value of an asset is constant over its useful life. This is illustrated in Figure 6.1 for an asset worth $1000 at the time of purchase and $200 eight years later. Graphically, the book value of the asset is determined by drawing a straight line between its first cost and its salvage or scrap value.

Figure 6.1 **Book Value Under Straight-Line Depreciation ($1000 Purchase and $200 Salvage Value After Eight Years)**



Straight-line method of depreciation

Algebraically, the assumption is that the rate of loss in asset value is constant and is based on its original cost and salvage value. This gives rise to a simple expression for the **depreciation charge per period**. We determine the depreciation per period from the asset's current value and its estimated salvage value at the end of its useful life, *N* periods from now, by

$$D = \frac{P - S}{N}$$

D the depreciation charge per unit time using the straight-line method

P the purchase price or current market value

S the salvage value after *N* periods

N the useful life of the asset, in periods

Computing Book Value

Similarly, the book value at the end of any particular period is easy to calculate: where BV_n the book value at the end of period n using straight-line depreciation.

$$BV_n = P - n\left[\frac{P - S}{N}\right]$$

A laser cutting machine was purchased four years ago for $380 000. It will have a salvage value of $30 000 two years from now. If we believe a constant rate of depreciation is a reasonable means of determining book value, what is its current book value?

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$$BV_n = P - n\left[\frac{P - S}{N}\right]$$

question 5

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From Equation (6.2), with $P = \$380\,000$, $S = \$30\,000$, $N = 6$, and $n = 4$,

$$BV_{sl}(4) = 380\,000 - 4\left[\frac{380\,000 - 30\,000}{6}\right]$$

$$BV_{sl}(4) = 146\,667$$

The current book value for the cutting machine is \$146 667. ■

If an asset has a first cost of \$50,000 with a \$10,000 estimated salvage value after 5 years, (*a*) calculate the annual depreciation and (*b*) calculate the book value of the asset after each year, using straight line depreciation.

(*a*) The depreciation each year for 5 years can be found

$D_t = (B - S)/n = (50{,}000 - 10{,}000)/5 = \$8000$

(*b*) The book value after each year t is computed using l

$BV_t = B - t\,D_t$

BV_1 (at year 1) $= 50{,}000 - 1(8000) = \$42{,}000$

BV_2 (at year 2) $= 50{,}000 - 2(8000) = \$34{,}000$

BV_3 (at year 3) $= 50{,}000 - 3(8000) = \$26{,}000$

BV_4 (at year 4) $= 50{,}000 - 4(8000) = \$18{,}000$

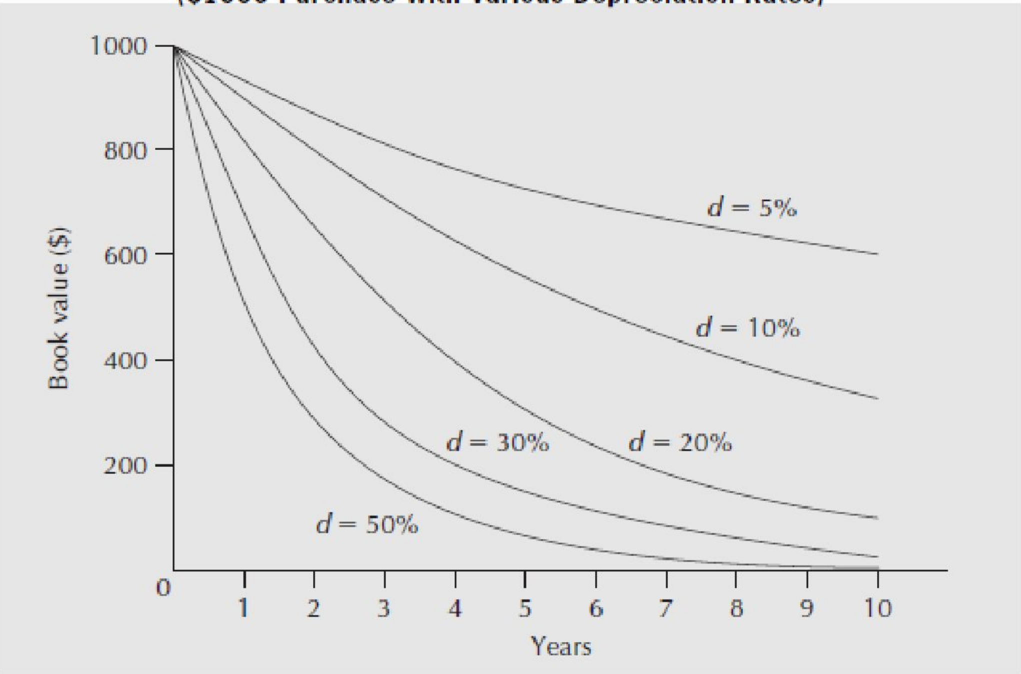
BV_5 (at year 5) $= 50{,}000 - 5(8000) = \$10{,}000$

Declining-Balance Depreciation

Declining-balance depreciation (also known as *reducing-balance depreciation*) models the loss in value of an asset over a period as a constant fraction of the asset's current book value. In other words, the depreciation charge in a particular period is a constant proportion (called the depreciation rate) of its closing book value from the previous period.

Depreciation is not a fixed amount each year
Instead, it is a constant fraction (rate) of the asset's book value at the start of the period

Figure 6.2 Book Value Under Declining-Balance Depreciation ($1000 Purchase With Various Depreciation Rates)



Declining-Balance Depreciation

Algebraically, the depreciation charge for period *n* is simply the depreciation rate multiplied by the book value from the end of period (*n* – 1).

$$D'_n = BV_{n-1} * d$$

- D'_n = The depreciation charge in period *n* using the declining-balance method
- BV_n = The book value at the end of period *n* using the declining-balance method
- D = The depreciation rate

The book value at the end of any particular period is easy to calculate by noting that the remaining value after each period is (1 - *d*) times the value at the end of the previous period.

$$BV_n = P - P*d$$

$$BV_n = P(1 - d)^n$$

Where, *P* the purchase price or current market value

Declining-Balance Depreciation

By using an asset's current value, *P*, and a salvage value, *S*, *n* periods from now, we can find the declining balance rate that relates *P* and *S*.

$$BV_n = S = P(1-d)^n$$

$$(1-d) = \sqrt[n]{\frac{S}{P}}$$

$$d = 1 - \sqrt[n]{\frac{S}{P}}$$

Paquita wants to estimate the scrap value of a smokehouse 20 years after purchase. She feels that the depreciation is best represented using the declining-balance method, but she doesn't know what depreciation rate to use. She observes that the purchase price of the smokehouse was \$245 000 three years ago, and an estimate of its current salvage value is \$180 000. What is a good estimate of the value of the smokehouse after 20 years?

$$d = 1 - \sqrt[n]{\frac{S}{P}}$$

$$BV_n = P(1 - d)^n$$

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$$d = 1 - \sqrt[n]{\frac{S}{P}}$$

$$= 1 - \sqrt[3]{\frac{180\,000}{245\,000}}$$

$$= 0.097663$$

$$d = 1 - \sqrt[n]{\frac{S}{P}}$$

$$BV_n = P(1 - d)^n$$

$$BV_{db}(20) = 245\,000(1 - 0.097663)^{20} = 31\,372$$

An estimate of the salvage value of the smokehouse after 20 years using the declining-balance method of depreciation is \$31 372. ■

Sherbrooke Data Services has purchased a new mass storage system for \$250 000. It is expected to last six years, with a \$10 000 salvage value. Using both the straight-line and declining-balance methods, determine the following:

(a) The depreciation charge in year 1

(b) The depreciation charge in year 6

(c) The book value at the end of year 4

(d) The accumulated depreciation at the end of year 4

$$D = \frac{P - S}{N}$$

Year	Depreciation Charge	Accumulated Depreciation	Book Value
0	-	0	\$ 250, 000
1	40,000		
2			
3			
4			
5			
6			10000

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(a) The depreciation charge in year 1
(b) The depreciation charge in year 6
(c) The book value at the end of year 4
(d) The accumulated depreciation at the end of year 4

Question 6

Year	Straight-Line Depreciation		
	Depreciation Charge	Accumulated Depreciation	Book Value
0			$250 000
1	$40 000	$ 40 000	210 000
2	40 000	80 000	170 000
3	40 000	120 000	130 000
4	40 000	160 000	90 000
5	40 000	200 000	50 000
6	40 000	240 000	10 000

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(a) The depreciation charge in year 1
(b) The depreciation charge in year 6
(c) The book value at the end of year 4
(d) The accumulated depreciation at the end of year 4

Year	Declining-Balance Depreciation		
	Depreciation Charge	Accumulated Depreciation	Book Value
0			$250 000
1	$103 799	$103 799	146 201
2	60 702	164 501	85 499
3	35 499	200 000	50 000
4	20 760	220 760	29 240
5	12 140	232 900	17 100
6	7 100	240 000	10 000

A three-year-old extruder used in making plastic yogurt cups has a current book value of $168 750. The declining-balance method of depreciation with a rate d 0.25 is used to determine depreciation charges. What was its original price? What will its book value be two years from now?

$$BV_{db}(3) = P(1 - d)^3$$

$$168\,750 = P(1 - 0.25)^3$$

$$P = 400\,000$$

The original price was $400 000.

$$BV_{db}(5) = 400\,000(1 - 0.25)^5 = 94\,921.88$$

or

$$BV_{db}(2) = 168\,750(1 - 0.25)^2 = 94\,921.88$$

The book value two years from now will be $94 921.88. ■

Practice Problem

(a) An asset costs \$14 000. What declining-balance depreciation rate would result in the scrap value of \$3000 after seven years?

(b) Using the depreciation rate from part (a), what is the book value of the asset after four years?

in notes